SAMANTHA SCOTT

Computer Engineering • Embedded Systems • C/C++ • Python • Linux
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PROFESSIONAL SUMMARY

Computer Engineering senior at Florida International University (GPA: 3.99), graduating December 2026 and recognized as the Fall 2026 Outstanding Bachelor's Graduate. Proficient in C/C++, Python, Linux, and embedded systems, with hands-on experience in hardware-software integration, ESP32, Raspberry Pi, BLE communication, sensor acquisition, and real-time monitoring. Experienced in system debugging, hardware validation, and technical troubleshooting.

TECHNICAL SKILLS

Programming : C, C++, Python, VHDL, Assembly, TypeScript, Java
Embedded Systems: ESP32, Raspberry Pi, Arduino, ARM Cortex-M, Zybo Z7 FPGA, sensor integration, firmware debugging
Communication & Protocols: BLE, GAP, GATT, I2C, UART, SPI, TCP/IP fundamentals
Linux & Development: Linux, Raspberry Pi OS, systemd, SSH, Git, POSIX
Hardware & Testing: Oscilloscope, logic analyzer, hardware validation, system integration, root-cause analysis

RELEVANT PROJECTS

DriveSense - Biometric Steering Wheel | *ESP32, Raspberry Pi, PPG Sensors, React Native, TypeScript, Supabase* **2026 – Present**

- Collaborate with a five-member engineering team to develop a biometric driver-monitoring system integrating six PPG sensors, ESP32, Raspberry Pi, and a mobile application for real-time physiological monitoring.
- Develop and integrate a React Native/TypeScript mobile application supporting BLE connectivity, real-time telemetry visualization, driver profiles, emergency contacts, and Supabase data synchronization.
- Implement **Sensi**, an AI-assisted safety system featuring on-device language-model inference, offline speech recognition, voice responses, and event-triggered emergency assistance.
- Test hardware-software communication, sensor acquisition, and safety workflows to validate system reliability and emergency-response functionality.

Raspberry Pi Linux Server Monitor | *Python, Linux, Raspberry Pi, systemd, SSH, UFW* **Sep 2026**

- Developed and deployed a Python-based monitoring utility on Raspberry Pi to track **5 system-health metrics**: CPU usage, memory, disk utilization, temperature, and uptime.
- Implemented configurable warning thresholds and timestamped logs using Linux /proc and /sys interfaces to monitor system health and identify abnormal conditions.
- Automated system monitoring at **5-minute intervals** using a systemd service and timer; configured SSH and UFW firewall rules for secure remote administration.

Traffic Light Controller with Pedestrian Interface | *Zybo Z7 FPGA, VHDL, C, Vivado/Vitis* **Apr 2026**

- Designed and implemented a finite-state machine (FSM) in VHDL to control traffic-light sequencing, pedestrian requests, and timed transitions, integrating GPIO inputs and OLED feedback.
- Synthesized and deployed the controller onto Zybo Z7 FPGA hardware using Vivado/Vitis; performed hardware testing to validate state transitions, pedestrian-request handling, and real-time operation.

Cross-Platform File Management & Locking Utility | *C++, POSIX, Windows API* **July 2026**

- Developed a cross-platform C++ utility supporting **Windows and Linux**, implementing direct-access file operations and shared/exclusive byte-range locking using POSIX fcntl() and Windows LockFileEx() APIs.

PROFESSIONAL EXPERIENCE

Lead Student Assistant – Housing IT | *Florida International University | Miami, FL* **Aug 2026 – Present**

- Led the replacement and validation of **200** electronic access-control devices across two residential buildings, coordinating installation, hardware testing, troubleshooting, and deployment tracking.
- Supervise and mentor **three** student assistants, assigning technical tasks, troubleshooting complex issues, and coordinating escalations with university staff and external vendors.
- Diagnose hardware, firmware, and connectivity failures in network-connected access-control systems, including wireless gateways, electronic locks, and device controllers.

Student Assistant – Housing IT | *Florida International University | Miami, FL* **Oct 2024 – Aug 2026**

- Troubleshot network-connected access-control systems across **9 residential buildings**, performing firmware updates, device configuration, and connectivity diagnostics on electronic locks, controllers, and wireless gateways.
- Supported **200+ endpoints** using Active Directory, SCCM, and JAMF; performed BIOS updates, driver installations, and hardware/software troubleshooting.

Student Assistant – Dean of Student Services | *Miami Dade College | Miami, Fl* **Nov 2022 – Apr 2024**

- Coordinated student records, official correspondence, and administrative operations while maintaining accuracy and meeting deadlines.

Customer Service Officer / Head Teller | *Bank of Nova Scotia Jamaica Ltd | Jamaica* **Feb 2016 – Aug 2022**

- Promoted to Head Teller; trained and guided **4 employees** while overseeing daily banking operations, regulatory compliance, and transaction accuracy.

EDUCATION

Florida International University | *Miami, FL* **Expected Dec 2026**
B.S. in Computer Engineering | GPA: 3.99/4.00
Honors & Awards: Outstanding Bachelor's Graduate Award (Fall 2026), Dean's List
Relevant Coursework: Embedded Computing, C++ Programming for Embedded Systems, Microcomputers I, Data Structures, Systems Programming, Operating System Principles, Python Programming in ECE, Telecommunication Networks, Circuit Analysis, Logic Design